RAJALAKSHMI ENGINEERING COLLEGE

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K2

Started on	Monday, 24 August 2026, 7:37 PM
State	Finished
Completed on	Monday, 24 August 2026, 7:39 PM
Time taken	2 mins 26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count = 0;
    int rev = 0, remainder;
    count++; // For rev = 0 assignment operation

    while (n != 0) {
        count++; // For the while condition check (true case)
        remainder = n % 10;
        count++; // For remainder assignment
        rev = rev * 10 + remainder;
        count++; // For rev assignment
        n /= 10;
        count++; // For n division assignment
    }
    count++; // For the final while condition check (false case)

    count++; // For the print(rev) statement from the original algorithm

    printf("%d", count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔

	Input	Expected	Got	
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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